

Fintech Payment Analytics Dashboard

Analysis of 6.3M Transactions | Fraud Detection & Pattern Intelligence

Total Transactions

6.36M

Total Fraud Cases

8197

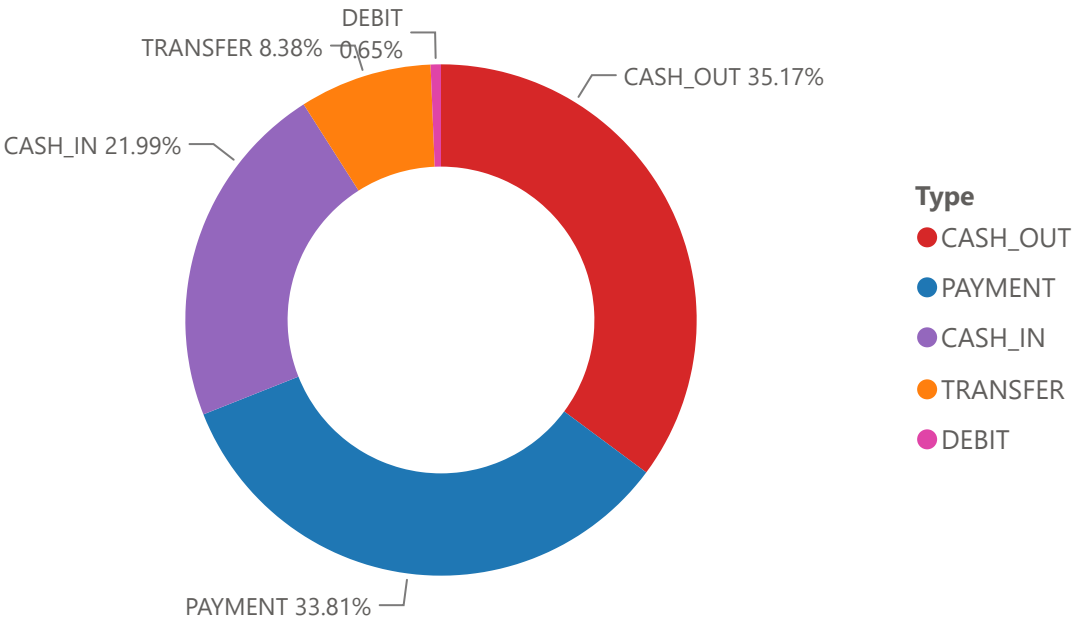
Fraud Rate (%)

12.88%

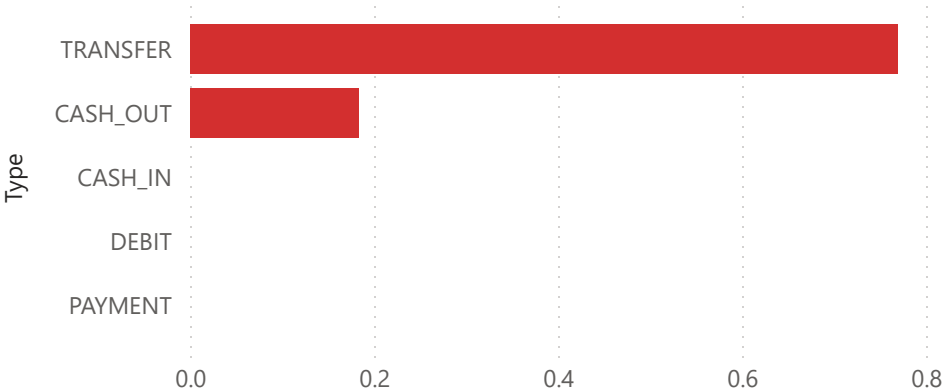
Total Fraud Value

\$12.06bn

Transaction Volume by Type



Fraud Rate by Transaction Type (%)



Key Findings:

Analysed 6.36M payment transactions identifying 8,197 fraud cases with a [correct %] fraud rate representing \$12.06bn in total fraud value. Fraud is exclusively concentrated in TRANSFER and CASH_OUT transaction types — PAYMENT, DEBIT, and CASH_IN recorded zero fraud cases.

Fraud Pattern Deep Dive

Top 30 Highest Risk Transaction Patterns

Type	Transaction Size	Time	Total Txns	Fraud Count	Fraud Rate (%)	Fraud Value	Avg Amount
TRANSFER	Very Large	Night	1974	341	17.27	1,270.0776M	25,69,019.90
TRANSFER	Medium	Night	1680	174	10.36	8.7618M	55,441.04
TRANSFER	Large	Night	5972	449	7.52	199.0132M	4,00,894.81
CASH_OUT	Large	Night	11714	451	3.85	199.5621M	2,47,498.36
CASH_OUT	Medium	Night	5573	174	3.12	8.7618M	55,297.26
TRANSFER	Medium	Morning	8500	186	2.19	9.0707M	54,690.78
TRANSFER	Small	Afternoon	2417	32	1.32	0.1269M	5,048.12
TRANSFER	Small	Evening	2505	33	1.32	0.1710M	4,971.51
TRANSFER	Very Large	Evening	32255	342	1.06	1,396.9829M	21,85,562.61
TRANSFER	Very Large	Morning	34983	338	0.97	1,246.4466M	22,52,874.82
TRANSFER	Medium	Afternoon	22015	188	0.85	9.4519M	55,240.39

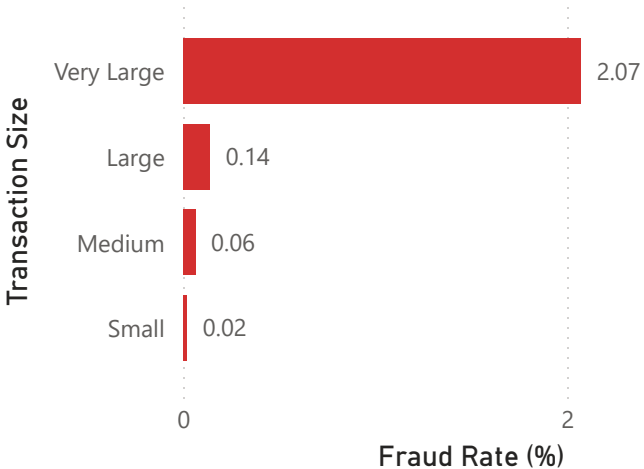
Key Findings:

- Fraud risk increased significantly with transaction size, with Very Large transactions recording the highest fraud rate (2.07%), making them the primary contributor to financial exposure.
- **TRANSFER** transactions during the Night period exhibited the highest fraud rate (35.15%), identifying this combination as the highest-risk transaction pattern.
- Fraud activity was concentrated in **TRANSFER** and **CASH_OUT** transactions, while **PAYMENT** and **CASH_IN** recorded negligible or no fraud.
- These findings suggest prioritizing real-time monitoring and enhanced fraud controls for high-value **TRANSFER** and **CASH_OUT** transactions, particularly during nighttime hours.

Fraud Rate Heatmap - Type vs Time of Day

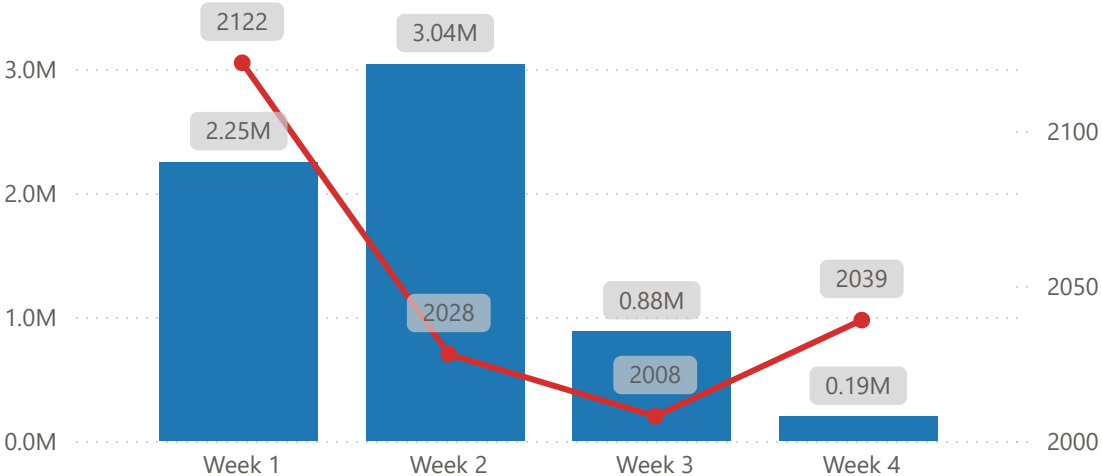
Type	Afternoon	Evening	Morning	Night
TRANSFER	3.0679	3.4346	3.9952	35.1501
PAYMENT				0.0000
CASH_OUT	0.2442	0.2657	0.6037	6.9723
CASH_IN	0.0000	0.0000	0.0000	0.0000

Fraud Rate by Transaction Size (%)

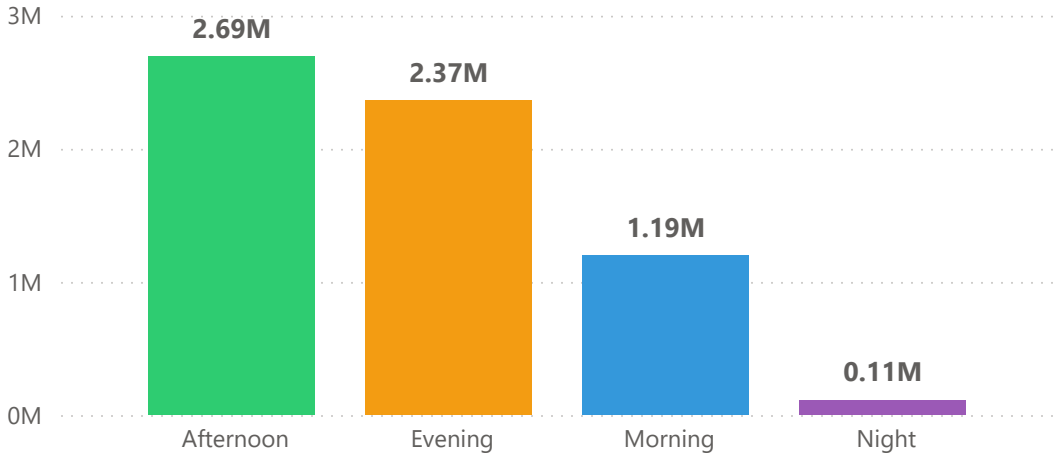


Transaction Volume & Trend Analysis

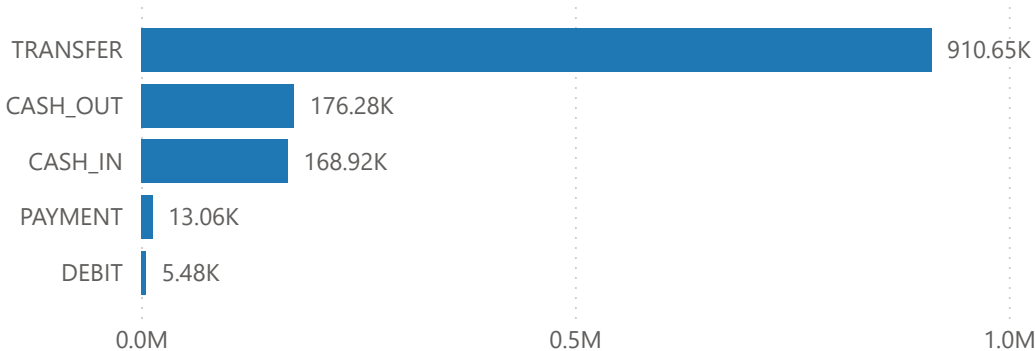
Weekly Transaction Volume vs Fraud Count



Transaction Volume by Time of Day



Average Transaction Value by Type



Key Findings:

- Transaction volume peaked in Week 2 with approximately **3.04 million** transactions before declining in Weeks 3 and 4.
- Afternoon recorded the highest transaction volume (**2.69M**), followed by Evening (**2.37M**), while Night had the lowest activity (**0.11M**).
- TRANSFER transactions had the highest average transaction value (**~910.65K**), indicating that high-value transfers represent the greatest financial exposure and warrant enhanced monitoring.

Fraud Detection System Analysis

Fraud Correctly Flagged

16

Fraud Missed by System

8181

False Positive

0

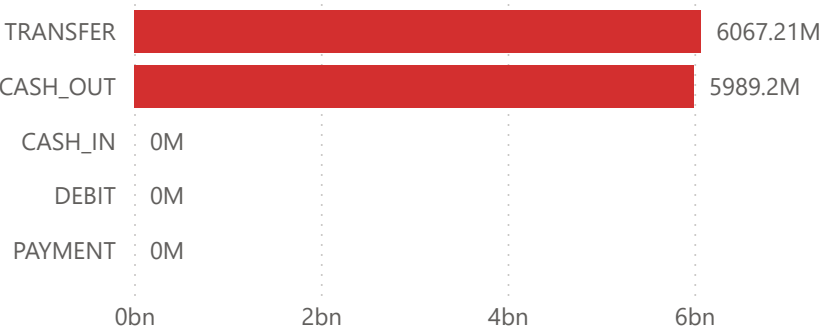
System Detection Rate

0.1952%

Fraud Detection System Performance

System Flag	Actual Flag	Transactions	Total Transaction
2	1	6.36M	11,44,392.94M

Total Fraud Value by Transaction Type



Critical Findings:

- The fraud detection system correctly identified only 16 of 8,197 fraudulent transactions, resulting in a **0.20%** detection rate.
- No false positives were generated, indicating high precision but extremely low fraud detection coverage.
- **8,181** fraudulent transactions remained undetected, highlighting a significant weakness in the current fraud detection mechanism.
- Strengthening real-time monitoring and deploying advanced machine learning models could substantially improve detection performance.